

COMPANION TO

Capability Matters: A Casebook

LENS Companion

CONCENTRATION · CROSSWALKS · REFERENCES

Learning Engineering for Next-Generation Systems · v2.1 (June 2026)

SECTION

How to use this companion

The LENS companion is a single, white-paper reference that pairs the program's concentration documentation with the casebook's crosswalks and indexes. It is built from the same source as **Capability Matters: A Casebook** but renders no case bodies — the casebook is the place to read the cases themselves; the companion is the place to find them.

The companion is in five parts:

- I. The Five LENS Competencies — formal names, taglines, main objectives, and the v2.1 subobjective set (1.1 – 5.6). The organising layer over the formal CLOs.
- II. The CLOs and the Course Mapping — the five concentration learning objectives, the LDT / LENS course descriptions, and the course-to-CLO coverage matrix.
- III. Crosswalks — the casebook's induced framework mapped to the canonical five competencies, and the LECF crosswalk to the IEEE ICICLE community work.
- IV. Casebook indexes — every case by primary domain, every case by LENS course, and the CLO + course coverage appendix.
- V. References by case — the appendix the casebook ships with, so a reader can find every source the volume draws on.

On versioning. This companion documents the **v2.1** program state (June 2026), in which D3 is renamed **Human-System Collaboration** and moved from position 5; T&E moves to D4; Sociotechnical Constraints moves to D5. The order now reads as the flywheel: see the system → build → integrate humans → measure → deploy. Prior versions are preserved in the program docs' change logs (`lens_program/1_LENS_Five_Competencies.md`).

SECTION

Why a companion — the thesis in brief

Learning engineering is a toolbox, not a single tool. At its centre is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design.

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

01. Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.
02. We engineer every parameter of a system except the people.
03. Capability is not a soft problem. It is a systems engineering problem.
04. Decision-grade evidence. Not opinions about training.
05. Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
06. Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver. When that interface is wrong, the gap may originate in human development, in system design, in organisational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialisation that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organised to help build it.

The casebook's method, in one sentence. Each case is an **unpacking**: take a real outcome, recover the capability question behind it, and attribute the gap to human development, system design, or organisational performance. The discipline of unpacking is what LENS teaches; the casebook is its primary instrument. The framework that follows is the analytic lens the unpacking method uses — five competencies, each a way of asking what the work the system requires actually is, and whether the people inside it can do it.

SECTION

The Five LENS Competencies — v2.1

Organising layer over the formal CLOs. Formal names carry program documentation; taglines carry recruiting and the site. Subobjective numbers (1.1, 3.2, ...) are stable tags for case studies, course modules, and capstone rubrics. The CLO language in the program documentation remains the language of record.

1 Systems Analysis

SYSTEMS THINKING AND ANALYSIS (CLO-1)

see the whole system

Main objective. Trace how human capability and system performance depend on each other, so interventions target the real problem.

- 1.1 Decompose system performance requirements into measurable human capability requirements: define what the system demands of its operators and the impact it must deliver.
- 1.2 Model learning environments and their host operational systems as interacting systems, with defined components, interfaces, and feedback loops.
- 1.3 Apply systems engineering lifecycle models to scope, sequence, and evaluate capability interventions within and across disparate systems.
- 1.4 Analyze and communicate human–system interdependencies to identify capability gaps and predict operational impact at scale.
- 1.5 **Governance–objection diagnostic.** Distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy. [v2]

2 Iterative Development

LEARNING ENGINEERING DESIGN AND IMPLEMENTATION (CLO-2)

build, test, refine

Main objective. Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

- 2.1 Design interventions that integrate learning sciences principles with measurable outcomes and system design constraints.
- 2.2 Run the iteration cycle: design, instrument, evaluate, refine, and revise the problem framing as evidence accumulates.
- 2.3 Evaluate evidence–based design strategies for transfer to high–consequence settings at operational speed and scale.
- 2.4 Construct implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

- 2.5 Narrate and defend the design iteration in first person.** Design competence includes not only producing an outcome but rendering the iteration legible. [v2]

3 Human-System Collaboration

HUMAN-SYSTEM COLLABORATION AND ADAPTIVE SYSTEMS (CLO-3)

work together

Main objective. Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system.

- 3.1** Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary.
- 3.2** Evaluate the measured impact of system-mediated work on human performance: the gains the system enables and the risks it introduces — automation bias, cognitive offloading, skill atrophy.
- 3.3 Delegation with revocation.** Design the human oversight layer for delegated work; specify in advance the disconfirming evidence that would revoke the delegation. [v2]
- 3.4 Collaboration measurement.** Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component. [v2]
- 3.5** Specify learning and capability requirements for systems not yet fielded, working from design artifacts rather than operational experience.

4 Test and Evaluation

DATA, MEASUREMENT, AND EVALUATION (CLO-4)

show what works

Main objective. Produce decision-grade evidence linking learning to operational impact — where decision-grade is a sufficiency judgment under irreducible uncertainty — and tell a learning gap from a system gap.

- 4.1** Design ethical instrumentation strategies that measure capability at the speed and scale operations require.
- 4.2 Gap attribution.** Diagnose the gap: differentiate capability shortfalls rooted in human development from those rooted in system design or organizational performance.
- 4.3** Construct decision-grade evidence artifacts under irreducible uncertainty: state what is known, what is assumed, and what would change the decision.
- 4.4 Judgment under inadequate evidence.** Justify a consequential decision on incomplete and contested evidence; document the basis; name what would change it. [v2]
- 4.5** Communicate evidence honestly to technical and non-technical stakeholders, including uncertainty and limits of inference.
- 4.6 Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence. [v2]

5 Navigating Sociotechnical Constraints

CONTEXT AND DOMAIN FLUENCY (CLO-5)

make it work in the real world

Main objective. Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive.

- 5.1 Analyze the regulatory, organizational, cultural, and technical constraints that shape what can be built, deployed, and sustained in a given context.
- 5.2 Apply human systems integration frameworks to fit capability interventions to operational environments.
- 5.3 Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early.
- 5.4 Translate constraints and stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.
- 5.5 Anticipate adoption and sustainment barriers, and design interventions that survive them.
- 5.6 **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable. [v2]

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

SECTION

CLOs and Course Mapping – v2.1

The LDT M.Ed. is a 10-course, 30-credit, online/asynchronous master’s degree in the School of Education. LDT comprises three concentrations: Learning Experience Design (LXD, currently operating), AI Leadership in Education (AILE, launching August 2026), and Learning Engineering (LENS, launching August 2026). LENS graduates demonstrate competency across five domains; the CLOs below are the assessable form.

THE FIVE CLOS

CLO-1 · Systems Thinking and Analysis. Decompose system performance requirements into measurable human capability requirements. Apply systems engineering lifecycle models. Analyze interdependencies between human and system performance to identify gaps and predict operational impact at scale.

CLO-2 · Learning Engineering Design and Implementation. Design capability development interventions through iterative engineering cycles. Evaluate evidence-based design strategies for transfer to high-consequence settings. Construct and communicate implementation plans.

CLO-3 · Human-System Collaboration and Adaptive Systems. Design human-system partnerships that make people more capable while preserving human agency. Specify delegation with revocation. Measure collaboration as a unit of analysis.

CLO-4 · Data, Measurement, and Evaluation. Design ethical instrumentation. Construct decision-grade evidence under irreducible uncertainty. Apply gap attribution. Demonstrate fairness beyond omission.

CLO-5 · Context and Domain Fluency. Integrate capability interventions into regulatory, organizational, cultural, and technical realities. Synthesize across boundaries. Anticipate adoption and sustainment barriers, including cross-regime / platform-dependency governance.

COURSE-TO-CLO COVERAGE

Course	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
LDT F1 (Learning Sciences Studio)		x			
LDT F2 (Critical Perspectives)					x
LEN 1 (Principles of LE for Complex Systems)	x	x			
LEN 2 (Human-AI Teaming and Adaptive Learning)			x		
LEN 3 (Learning Engineering Systems)	x				x
LEN 4 (Evidence, Analytics, Measurement)				x	
LEN 6 (Applied Problem Solving in LE)	x	x			x
LEN 10 (Capstone)	x	x	x	x	x
LEN 5 (Human Capability Analysis, elec.)	o				o
LEN 7 (Bias, Risk, Governance, elec.)					o

LEN 8 (Knowledge Transfer & Org Learning, elec.)		○			○
LEN 9 (Computational and AI Methods, elec.)			○	○	

✕ = required course; ○ = elective. Foundations + concentration + methods + capstone = 8 required courses (24 credits); 2 electives complete the 30-credit degree.

SECTION

Crosswalks

INDUCED 8 (CASEBOOK SCAFFOLD) → CANONICAL 5 (PROGRAM OF RECORD)

The casebook induces an 8-cluster analytic scaffold from its 194 cases (full account: `competencies.md`). Those eight cluster cleanly into the canonical five LENS competencies — except for D2 Iterative Development, which is the iteration **method** threaded through the cases rather than its own cluster. The gap is what the editor-commissioned first-person Practice Flywheel accounts will close in the next edition.

Induced cluster (casebook)	Canonical D	Notes
1 Capability requirements under operational reality	D1	Engineered vs. stated requirements; sustainment-as-requirement.
2 Evidence architecture the institution cannot deceive itself with	D4	Measurement gaming; post-deployment surveillance.
3 Interface and role design at the human-automation boundary	D3	Cues, alerts, mode/state transparency, recoverability.
4 Pairing mechanism with authorization	D3	Frontline halt authority; non-punitive reporting; authority gradient.
5 Governance architecture for deployment	D5	Consent, oversight, accountability as design artifacts.
6 Cross-organization and cross-time knowledge transfer	D5	Institution-building after catastrophe; tacit-knowledge survival.
7 Capability under system change and aging assumptions	D5	Legacy assets; multi-layer drift; transitions.
8 Equity and construct definition as design commitments	D4	Construct choice; demographic stratification; fairness.
(none specific)	D2	The iteration method — threaded across cases, not its own cluster.

THE CASEBOOK'S THREE ANCHORS PER CASE

Every v2 case carries three anchor codes in its header and metadata:

- induced-anchor** — e.g. "2.4" — the casebook's induced 8 / 32 sub-cluster the case evidences (analytic scaffold).
- lens-anchor** — e.g. "D3/PT5" — the canonical LENS five domain + problem type (1 – 6) the case is **primary** for.

clo-anchor — e.g. "CL0-3" — the assessable CLO the case serves as a worked example for (course mapping anchor).

SECTION

Casebook indexes

The same indexes that ship with **Capability Matters: A Casebook**, in v2.1 form. They are queried from the casebook source so they stay in sync as cases are added or revised.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE

Case 5

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater–Nichols.

STANDOUT SUCCESS

Case 18

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability–engineering program in any high–consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

1	USS Fitzgerald & USS John S. McCain	2017	T·K·N
2	Marine Corps Training in the INDOPACOM AOR	ongoing	T·K
3	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
4	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
5	Operation Eagle Claw	1980	T·K
8	Patriot Missile / Dhahran	1991	D·H·K
17	GAO Weapon–System Sustainment Reviews — Operating Costs without Decision–Grade Data	2022 (GAO–22–104533)	G·K
18	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
21	MIL–STD–1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K
33	GIFT and the Adoption Gap	2012 – present	K·G·N

37	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N
42	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D
54	USS Vincennes & Iran Air Flight 655	1988	H·T
64	Stanislav Petrov / 1983 False Alert	1983	H·T
138	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
141	V-22 Osprey	1991 – present	T·H·N
167	9/11 Intelligence Sharing Failures	1996 – 2001	G·K

AVIATION

STANDOUT FAILURE

Case 7

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 70

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

6	AeroPerú Flight 603	1996	H·T
7	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
12	Boeing 737 Rudder Hardovers	1991, 1994	H·D
28	Air France Flight 447	2009	T·H
55	Kegworth / British Midland 92	1989	T·H·K
56	Asiana Airlines Flight 214	2013	T·H
57	Helios Airways Flight 522	2005	T·H
58	TransAsia Airways Flight 235	2015	T·H
63	Eastern Air Lines Flight 401	1972	H·T
68	Air Canada Chatbot Liability — Delegation Without Revocation	2022 – 2024	D·K·N
70	Crew Resource Management & CAST	1981 – present	T·H·N
72	Korean Air Safety Transformation	2000 – present	T·N
82	Colgan Air Flight 3407	2009	T

83	Atlas Air Flight 3591	2019	T·K
88	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N
111	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
118	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
119	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson Version 7.1)	1981 – 2008 (TCAS II	H·K·G
164	NASA Saturn V Documentation	1972 – present	K
175	Singapore Airlines Safety Transformation	1980s – present	T·N

HEALTHCARE

STANDOUT FAILURE

Case 61

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS

Case 109

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

20	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
32	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
34	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
59	Therac-25	1985 – 1987	H·D·G
61	EHR / CPOE Implementation	2005 – present	H·D·G
71	Keystone ICU / Pronovost Checklist	2004 – present	T·N
74	TREWS / Sepsis Watch	2018 – 2022	H·K·G
75	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H·K·N
79	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D
86	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G·N·K
93	Epic Sepsis Model	2017 – 2021	D·G·N
102	Medical Errors as Systemic Failure	1999 – present	T·H·N·K·G

105	Vioxx Withdrawal	1999 – 2004	G·D
109	WHO Surgical Safety Checklist	2008 – present	T·N
112	Bristol Heart Babies Reform	1984 – present	G·N
125	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
126	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N
129	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
130	Radiology AI Miscalibration	2018 – present	H·K·D
132	AlphaFold — Protein Structure Prediction	2020 – present	T
136	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	T·K·D
150	Theranos	2003 – 2018	G·D
173	TeamSTEPPS	2006 – present	T·N
174	Tylenol Recall	1982	G·N

EDUCATION AT SCALE

STANDOUT FAILURE

Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS

Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

38	Cognitive Tutor / Carnegie Learning	1990s – present	T
89	Tennessee Voluntary Pre-K Study	2009–2018	G·D
90	Algorithmic Bias in Educational Predictive Analytics	ongoing	G·H·D
104	Atlanta Public Schools Cheating Scandal	2009 – 2015	G·N
110	Georgia State University Predictive Analytics	2012 – present	T·K
146	inBloom	2014	G
149	Summit Learning / Personalized Learning Rollout	2014–2019	G·T·K

171	xAPI / Total Learning Architecture — Interoperability Gap	ongoing	K·G
194	The Discipline We Build Next	ongoing	T·K·N·G·H·D

ENERGY

STANDOUT FAILURE	Case 69	Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.	
	Case 172	INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.	
11	Sago Mine Disaster	2006	N-T-K
60	Northeast Blackout	2003	H-K
69	Three Mile Island	1979	T-H
85	Deepwater Horizon	2010	T-N-K
87	Upper Big Branch Mine Explosion	2010	N-G-K
137	Texas City BP Refinery Explosion	2005	N-T-K-G
142	Davis-Besse Reactor Head Corrosion	2002	N-K-G
143	Fukushima Daiichi	2011	N-G-K
154	Camp Fire / PG&E	2018	G-N-K
172	INPO and the Nuclear Academy	1979 – present	T-K-G

GOVERNMENT

STANDOUT FAILURE	Case 162	Australia Robodebt (2016–20). An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.
NOTE	Case 111	No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

14	Healthcare.gov Launch	2013	K·T·G
101	VA Wait-Time Scandal	2014	G·K·N
103	LIBOR Manipulation	2003 – 2012	G·N
131	Predictive Policing — PredPol	2011 – present	G·H·D
166	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N

INDUSTRIAL

STANDOUT FAILURE	Case 147	Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.
STANDOUT SUCCESS	Case 73	Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

30	Kodak Digital Camera Stagnation	1975–2012	D·K·N
73	Toyota Production System / Andon Cord	1950s – present	N·G
84	Takata Airbag Inflators	2008 – 2023	D·G
92	Volkswagen Dieselgate	2015	D·G
139	GM Ignition Switch	2002 – 2014	D·G
147	Bhopal	1984	T·K·N·G
148	Grenfell Tower	2017	G·T·K·N
153	Hyatt Regency Walkway Collapse	1981	D·G

TECHNOLOGY

STANDOUT FAILURE	Case 66	UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters’ reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.
NOTE	Case 132	The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 132), the contemporary worked example of computational capability-engineering done as a discipline.

10	Knight Capital Trading Loss	2012	D·K
31	BlackBerry Touchscreen Inertia	2007–2013	D·N·K
66	UK Post Office Horizon Scandal	1999 – 2015	G·H·K
80	AI-Augmented Coding Tools	2021 – present	T·H
91	Wells Fargo Fake Accounts	2011 – 2016	G·N
144	CrowdStrike Falcon Outage	2024	D·K·G
145	TSB Bank IT Migration	2018	H·G
151	Cambridge Analytica / Facebook	2014 – 2018	G
152	Equifax Data Breach	2017	G·K

AI IN EDUCATION

135	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N
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K–12 EDUCATION

36	Growth–Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
52	Chen et al. — Rural China AI Devices and the Equity–Direction Finding	2025	T·K·D
97	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G·K·N

K–12 MATHEMATICS

49	ASSISTments — National Replication and Long–Term Follow–Through	2014 – present	T·K·D
127	Cognitive Tutor Algebra I at Scale — Year–One Null, Year–Two Positive	2007 – 2014	T·K·D

K–12 SCIENCE

51	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

25	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change	2005	G·K·H
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ANESTHESIOLOGY

116	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AUTONOMOUS SYSTEMS

STANDOUT
FAILURE

Case 62

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE

Case 194

No paired-intervention success in the autonomous-systems dataset. The capability engineering this category needs is still being built — see Case 194 (**The Discipline We Build Next**) for the open question this volume closes on.

62	Uber ATG / Tempe Fatality	2018	T·N·G·H
78	Tesla Autopilot — Recurring Fatalities	2016 – present	T·N·G·H
192	Cruise Robotaxi — Pedestrian Drag	2023	G·D·H

AUTONOMOUS VEHICLES

156	Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment	2023	G·K·N
183	Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact	2023 / 2025	G·K·N
184	CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver	2020 – 2024	G·K·D

AVIATION INFRASTRUCTURE

23	FAA NextGen and the ADS-B Out Transition	2003 – 2020	G·D·K
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AVIATION SAFETY

22	FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule	1988 – 2010s	G·D·K
180	FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike	1992 – 2000s	G·K·N

CLINICAL CARE

43	I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer	2014	H·K·N
120	Bar-Code Medication Administration — Cue at the Point of Care	2010	H·K·D

CLINICAL MEDICINE

95 Racial Bias in Pain Assessment — The False-Belief Mechanism 2016 **T·K·N**

113 Removing the Race Coefficient from eGFR 2021 **D·G·N**

CLINICAL/TRANSLATIONAL RESEARCH

178 Team Science Training for Clinical and Translational Scientists 2020 – 2022 **K·N**

CORPORATE L&D

35 Training Transfer — The Gap Between Learning and Doing 2010 **K·N**

46 High-Impact Learning System — Engineering the Environment, Not Just the Event 2001 – present **K·N**

108 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present **K·N**

124 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2005 – present **K·N**

CRIMINAL JUSTICE

99 COMPAS Recidivism Prediction — Calibration vs. Equal Error Rate 2014 – 2018 **D·K·N**

DATA GOVERNANCE

190 CARE Principles — Indigenous Data Governance as a Replaced Regime 2019 – 2020 (principles); ongoing **G·N**

DIGITAL IDENTITY

187 Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India 2018 – 2025 **G·N·H**

E-GOVERNMENT

27 Estonia X-Road — Continuous Migration as a Governance Pattern (and the No-Legacy Paradox) 2001 – present **G·K·N**

ED-TECH

96 Algorithmic Bias in Automated Exam Proctoring 2022 **D·N·K**

EDUCATION (NORWAY)

182 Norway's National Expert Commission on Learning Analytics 2022 – 2023 **G·K·N**

EDUCATION AT SCALE

40 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T·D**

100 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2024 **D·K·N**

FINANCE

94 Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity 2018 – 2022 **D·G·N**

107 Fintech Lending Fairness Audit — When Including Race Reduces Disparity 2025 **D·G·N**

FINANCIAL SERVICES

163 Apple Card / Goldman Sachs — When the Lender Cannot Explain Its Own Model 2019 – 2021 **D·K·N**

GENERAL AVIATION

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No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 18) for the safety-engineering counterpart that aerospace did not build.

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Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

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- 122 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units
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- 123 MMALA — A Maturity Model for Responsible Learning–Analytics Adoption (Brazil)
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- 124 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument
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- 127 Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive
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- 128 OU Analyse — Predictive Learning Analytics and Teacher Use at Scale
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- 138 Mark 14 Torpedo Failures
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- 143 Fukushima Daiichi
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- 145 TSB Bank IT Migration
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- 151 Cambridge Analytica / Facebook
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- 3 F–35 Sustainment & Maintainer Shortage
- 4 Military Fratricide — Desert Storm to Afghanistan
- 5 Operation Eagle Claw
- 8 Patriot Missile / Dhahran
- 9 Mars Climate Orbiter — Unit Mismatch
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- 16 EU Human Brain Project — Top–Down Vision That Unraveled
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10	Knight Capital Trading Loss

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 - 38** Cognitive Tutor / Carnegie Learning

 - 61** EHR / CPOE Implementation

 - 67** Watson for Oncology — Delegation Marketed Ahead of Capability

 - 68** Air Canada Chatbot Liability — Delegation Without Revocation

 - 90** Algorithmic Bias in Educational Predictive Analytics

 - 94** Algorithmic Mortgage Lending — Omitting the Variable Did Not Fix the Disparity

 - 96** Algorithmic Bias in Automated Exam Proctoring

 - 107** Fintech Lending Fairness Audit — When Including Race Reduces Disparity

 - 113** Removing the Race Coefficient from eGFR

 - 114** Pulse Oximetry Across Skin Tones

 - 129** COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

 - 130** Radiology AI Miscalibration

 - 131** Predictive Policing — PredPol

 - 132** AlphaFold — Protein Structure Prediction

 - 133** Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

 - 134** Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

 - 136** DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback

 - 155** Dutch SyRI — Welfare-Fraud Risk Scoring Halted on Rights Grounds

 - 156** Cruise's Partial Disclosure — How Disclosure Posture Decides Deployment

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 - 183** Waymo's Safety Case Framework — Governance Objection Dissolved by Designed Artifact

 - 184** CPUC AV Passenger-Service Permits — Conditions as a Designed Objection-Dissolver

 - 187** Aadhaar Exclusion Litigation — Judicial Correction of Biometric Welfare Delegation in India

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The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Human-System Collaboration	CLO-3 · Human-System Collaboration and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49
LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86

LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

References, case by case.

Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in casebook/verification-log.md. Future editions will fill in remaining locators as the per-case review closes them out.

Style. References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

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2017 – present

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2008 – 2023

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CASE 90 Algorithmic Bias in Educational Predictive Analytics

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CASE 97

School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism

2010s – 2022

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UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days

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2014 – 2018

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1999 – present

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CASE 103 LIBOR Manipulation

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2009 – 2015

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CASE 105 Vioxx Withdrawal

1999 – 2004

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CASE 106 Purdue Course Signals — The Reverse-Causality Retention Claim

2012 – 2013

1. Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. Proceedings of LAK 2012, 267–270. doi:10.1145/2330601.2330666 — the original study at the center of the critique.

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CASE 107 Fintech Lending Fairness Audit — When Including Race Reduces Disparity

2025

1. Coots, Bartlett, Nyarko, & Goel (2025), "Algorithmic Bias in Lending: Evidence from a Fintech Audit," arXiv:2512.20753 — audit of 80,000 personal loans showing model miscalibration disparities and that controlled inclusion of protected attributes could correct them; the evidence-tier flag is binding until peer review completes.

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CASE 108 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present

1. Kirkpatrick (1959–1960), original four-level evaluation series in *Journal of the ASTD*; updated as Kirkpatrick & Kirkpatrick, *Evaluating Training Programs* (3rd ed., 2006).

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3. D2L, "Kirkpatrick's 4 Levels of Training Evaluation," practitioner guide documenting the same pattern.

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CASE 109 WHO Surgical Safety Checklist 2008 – present

1. Haynes, A. et al. (2009), "A Surgical Safety Checklist to Reduce Morbidity and Mortality," *NEJM* — the 1.5%→0.8% result and major-complication reduction.

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CASE 110 Georgia State University Predictive Analytics

2012 – present

1. Renick, T. & Strom, A. (2020) on GSU’s advising transformation — the system design and outcomes.

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2. Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data.

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5. Complete College America, Game Changers documentation — dissemination of the model.

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CASE 111 Aviation Safety Reporting System (ASRS)

1976 – present

1. NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume.

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CASE 112 Bristol Heart Babies Reform

1984 – present

1. Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children's Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and recommendations (paraphrased).

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CASE 113 Removing the Race Coefficient from eGFR

2021

1. Delgado et al. (2021), "A Unifying Approach for GFR Estimation: Recommendations of the NKF-ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease," American Journal of Kidney Diseases 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online-first year, the year of the Task Force recommendation.

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CASE 114 Pulse Oximetry Across Skin Tones

1990 – 2020 – 2025

1. Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), "Racial Bias in Pulse Oximetry Measurement," New England Journal of Medicine 383(25):2477–2478, doi:10.1056/NEJMc2029240.

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3. FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff," issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540 — the regulatory corrective-action artifact, language may evolve in final.

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CASE 115 Open University 'Ethical Use of Student Data' and OU Analyse

2014 – 2025

1. Slade & Prinsloo (2013), "Learning Analytics: Ethical Issues and Dilemmas," American Behavioral Scientist 57(10):1510–1529, doi:10.1177/0002764213479366.

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CASE 116 Anesthesia Monitoring Standards and the APSF — The Mortality Decline

1986 – present

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CASE 117 Interprofessional Education and the Evidence Gap

2013 – 2015

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CASE 118 EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation 1996 – 2002

1. Bateman, C. D. (1999), “The Introduction of Enhanced Ground Proximity Warning Systems (EGPWS) into Civil Aviation Operations Around the World,” in Proceedings of the 11th Annual European Aviation Safety Seminar, Flight Safety Foundation, pp. 259–274 — developer history.

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CASE 119 **TCAS — Coordinated Collision Avoidance and the Überlingen Lesson** 1981 – 2008 (TCAS II Version 7.1)

1. RTCA (2008), DO-185B “Minimum Operational Performance Standards for Traffic Alert and Collision Avoidance System II (TCAS II)” — Version 7.1 with reversal logic.

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2. Bundesstelle für Flugunfalluntersuchung (BFU) (2004), AX001-1-2/02 — Investigation Report on the mid-air collision on 1 July 2002 near Überlingen.

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CASE 120 **Bar-Code Medication Administration — Cue at the Point of Care** 2010

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CASE 121 Surgical Skill Video Peer-Rating Predicts Complications

2013

1. Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. *New England Journal of Medicine*, 369(15):1434–1442. doi:10.1056/NEJMs1300625 — the case's primary evaluation.

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CASE 122 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

2009 – 2016

1. Vedula, Ishii, & Hager (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," *PLOS One* 11(3):e0149174, doi:10.1371/journal.pone.0149174.

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CASE 123 MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil)

2024

1. Freitas, Mello, Gasevic, Costa, & Andrade (2024), "MMALA: Developing and Evaluating a Maturity Model for Adopting Learning Analytics," *Journal of Learning Analytics* 11(1):67–86.

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3. Hilliger et al. (2020), Internet and Higher Education — LALA participatory adoption companion (Case 181).

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4. Norwegian Expert Commission on Learning Analytics, final NOU (2023) — national-scale companion (Case 182).

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CASE 124

Brinkerhoff Success Case Method — Tails as the Evaluation Instrument

2005 – present

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CASE 125

Annual-Screening UI Redesign + CDS at University of Missouri Health Care

2020

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4. Middleton et al. (2013), "Enhancing patient safety and quality of care by improving the usability of electronic health record systems," *JAMIA* 20(e1):e2–e8 — the framing peer-reviewed paper on EHR-usability-as-safety.

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CASE 126 **Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal** 2019 – 2024

1. Co et al. (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," JAMIA 26(10):1141, doi:10.1093/jamia/ocz095.

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2. Russ et al. and colleagues (2024), "Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization," PubMed 39049299 — nursing-workflow QI redesign of four high-firing alerts.

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3. Authors (2024), "Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support," Applied Clinical Informatics — interruptive-to-passive conversion with dual outcomes.

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CASE 127 **Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive** 2007 – 2014

1. Pane, J. F., Griffin, B. A., McCaffrey, D. F., & Karam, R. (2014), "Effectiveness of Cognitive Tutor Algebra I at Scale," Educational Evaluation and Policy Analysis 36(2):127–144, doi:10.3102/O162373713507480.

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4. What Works Clearinghouse — Cognitive Tutor evidence-base summary applying federal-grade evidence standards to the Pane et al. trial.

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CASE 128 **OU Analyse — Predictive Learning Analytics and Teacher Use at Scale** 2019 – 2023

1. Herodotou, C., Hlosta, M., Boroowa, A., Rienties, B., Zdrahal, Z., & Mangafa, C. (2019), "Empowering online teachers through predictive learning analytics," British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853.

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CASE 129 **COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case** 2022 – 2024

1. Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), "Impact of a deep learning sepsis prediction model on quality of care and survival," NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6.

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CASE 130 **Radiology AI Miscalibration** 2018 – present

1. Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), "Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis," PNAS 117(23):12592–12594 — sensitivity drops for under-represented groups on NIH ChestX-ray14 and CheXpert.

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3. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), "Dissecting racial bias in an algorithm used to manage the health of populations," *Science* 366(6464):447–453 — proxy-label bias in care-management algorithm covering 200 million Americans.

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5. FDA, "Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device" (2019); FDA draft guidance on AI/ML-Based SaMD (2025), with Predetermined Change Control Plan and Good Machine Learning Practice principles — the regulatory trajectory; absence of mandatory demographic stratification at clearance and post-market monitoring of in-use performance.

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CASE 131 Predictive Policing — PredPol

2011 – present

1. Lum, K. & Isaac, W. (2016), "To Predict and Serve?," *Significance* — the enforcement-vs-crime feedback loop (paraphrased).

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2. Richardson, Schultz & Crawford (2019), "Dirty Data, Bad Predictions" — biased records feeding predictive systems.

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CASE 132 AlphaFold — Protein Structure Prediction

2020 – present

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2. Varadi et al. (2022), AlphaFold Protein Structure Database, *Nucleic Acids Research* — the 200M+ structures and open release.

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3. Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased).

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4. CASP benchmark documentation — the decades-long agreed evaluation.

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5. Hassabis (DeepMind) public commentary — the open-release governance decision.

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CASE 133

Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction

2024

1. Gándara, Anahideh, Ison, & Picchiarini (2024), “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction,” AERA Open 10, doi:10.1177/23328584241258741 — primary case source on cross-group accuracy disparity, stratified evaluation, and bias mitigation in student-success prediction.

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4. v2 cross-referenced cases: 105 (eGFR race correction), 106 (pulse oximetry across skin tones), 107 (Hoffman pain bias) — the race-construct trio at the construct-definition layer.

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CASE 134

Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models

2021 – 2024

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